

Experiment - 02.

- AIM: Classification using Deep Neural Network (Any one from the following.)
 1. Multiclass classification using deep neural networks:
Example: Use the OCR letter recognition dataset.
 2. Binary classification using deep neural network example:
Classify movie reviews into "positive" reviews and "negative" reviews, just based on the text content of the reviews. Use IMDB dataset.

THEORY:

(1) Introduction-

- Classification is one of the most fundamental tasks in machine learning, where the objective is to assign input data into pre-defined categories or classes.
- Sentiment analysis is a classic example of a binary classification problem, where text data is classified into two categories: positive sentiment or negative sentiment.
- Movie review sentiment analysis is widely used by streaming platforms, review aggregators, and recommendation systems to understand audience opinions to improve

user experience.

(2). Dataset description (IMDB dataset) -

- The IMDB movies reviews dataset is a widely used benchmark dataset for sentiment analysis tasks.
- Dataset characteristics -
 - Total reviews : 50,000.
 - Training set : 25000 reviews.
 - Testing set : 25000 reviews.
 - Classes : Positive (1)
Negative (0)
 - Data type - Text.
 - Class Balance: Balanced dataset.

(3). Deep Neural Network for text classification -

- Text data is inherently unstructured and high dimensional. DNN offer several advantages -
 - Ability to learn complex patterns from text.
 - Automatic feature extraction.
 - Better performance compared to shallow models.
 - Scalability for large datasets.

(4) Text preprocessing -

- Before feeding text data into a deep Neural Network, it must be converted into a numerical format.

- Pre-processing steps:

1. Tokenisation - Splitting reviews into words.

2. Vocabulary limitation - Selecting most frequent words.

3. Integer encoding - Mapping words into numerical indices.

4. Padding - Ensuring equal input length for all reviews.

These steps transform variable length text reviews into fixed size numerical sequences suitable for neural network input.

(5). Model Architecture -

- The deep Neural Network used consists of multiple layers designed to extract meaningful features from text.

- Architecture Overview -

1. Embedding layer -

- Converts words into dense vector representations.

- Captures semantic relationships between words.

2. Fully connected (Dense) Hidden layers -

- Learn higher level abstract features.
- Use non linear activation functions (ReLU).

3. Output layer -

- Single neuron with Sigmoid activation.
- Produces a probability rate between 0 and 1.

(6) Training Configuration -

- 1) Loss Function. - Binary cross-entropy loss. - suitable for binary classification problems.
- 2) Optimizer - Adam optimizer - combines advantages of momentum and adaptive learning rates.
- 3) Evaluation metric - Accuracy - Measures the percentage of correctly classified reviews.

(7) Model Training.

- During training, model learns word representations and sentiment patterns.
- Weights are updated using back propagation.
- The loss gradually decreases as model improves.

(8). Results and observations -

- The trained DNN typically achieves high accuracy due to its ability to capture semantic and contextual patterns in text data.
- The model successfully distinguishes between positive and negative reviews.
- Reviews containing emotionally strong words influence predictions strongly.
- Longer reviews provide richer contextual information.
- The embedding layer significantly improves classification performance.

(9). Advantages of using deep Neural Networks -

- Automatic feature extraction from raw text.
- Better handling of large datasets.
- High prediction accuracy.
- Can be extended to multiclass sentiment analysis.
- Scalable to real world applications.

(10). Applications -

- Movie and product review analysis.
- Social media sentiment monitoring.
- Brand reputation management.
- Recommendation systems.
- Customer feedback analysis.

(11). CONCLUSION:

Thus Deep Neural Network was successfully implemented for binary classification of movie reviews using the IMDB dataset.

This study highlights the effectiveness of Deep Neural Network in handling unstructured text data and demonstrates their importance in modern NLP applications.

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